# LESSON PLAN

## LSF, STABILITY, A/C PERFORMANCE (6)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 19 |
| **Total No. of Lesson Plans** | 33 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Level flight performances, power available and required for jet and propeller aircrafts. |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Level flight performances, power available and required for jet and propeller aircrafts. Best climb speeds. |
| **5. Specific Objectives** | 6.1.3.1 Trainee will be able to describe flight performance of aircraft. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of power required vs power available curves for Jet and Propeller aircraft; Instructor supervision during curve analysis. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) Define level flight in terms of vertical forces.
(b) What is the relationship between thrust and drag in steady level flight?
(c) How does power required change with airspeed for a propeller aircraft?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Definition of Level Flight Performance<br>• Power Available Characteristics for Jet Aircraft<br>• Power Required Characteristics for Jet Aircraft<br>• Power Available Characteristics for Propeller Aircraft<br>• Power Required Characteristics for Propeller Aircraft | L & Ds | PPT, IB | 15 min |
| (b) | • Intersection of Power Curves for Best Level Speed<br>• Excess Power Concept in Level Flight<br>• Best Climb Speeds for Jet Aircraft<br>• Best Climb Speeds for Propeller Aircraft<br>• Effect of Weight on Level Flight Performance | L & Ds | PPT, PDF | 20 min |
| (c) | • Comparison of Jet vs Propeller Performance<br>• Operational Implications of Power Margins<br>• Summary of Level Flight Performance Parameters<br>• Review of Best Climb Speed Definitions<br>• Preparation for Forces in Climb Lesson | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain why best climb speed differs between jet and propeller aircraft.
(b) Describe the significance of the intersection point on the power required and available curves.
(c) How does an increase in aircraft weight affect the power required curve?

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| **12. Home Assignment and Follow Up** | Prepare a brief note comparing the power available characteristics of a turbojet engine versus a turboprop engine for the next lesson on climb performance. |

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| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Forces in climb, Excess power, max angle, maximum rate, Economical climb, time to climb. | Refer to the master syllabus reference list | LP 20 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min